# Systematic Literature Review Protocol

## Applications of Machine and Computer Vision in Software Engineering

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# 1 Introduction

This protocol defines the methodology for a Systematic Literature Review (SLR) investigating the applications of machine and computer vision in Software Engineering. With the increasing adoption of visual representations, diagrams, screenshots, logs, and interface artifacts throughout the Software Development Life Cycle (SDLC), computer vision techniques are increasingly being explored to automate, support, and enhance software engineering tasks.

It would be good to add some citations to the introduction

The objective of this SLR is to systematically identify and synthesize existing research that applies computer vision and generative AI techniques to software engineering problems. In particular, the review focuses on understanding which software engineering tasks are addressed using computer vision, the types of vision models employed, and the application areas in which generative AI is integrated with vision-based approaches. Additionally, this review examines the challenges encountered when applying computer vision in software engineering contexts and the performance metrics used to evaluate such approaches.

can be both text and vision models...

just to be clear computer vision manny domains, traditional and theoritical modes, ML models, DL models, VLMs. focusing on all?

By addressing the defined research questions (RQ1–RQ5), this review aims to provide a structured overview of the current state of research, highlight gaps in existing literature, and inform future research directions at the intersection of computer vision and software engineering.

# 2 Research Questions

The research questions need to be refined further. I suggest to refer back to previous drafts and discussions with Maleknaz.

The review is guided by the following research questions:

again computer vision is quite broad, need to be more focused

- RQ1: What software engineering tasks are addressed by computer vision?

To what extent computer vision models are integrated in SE state of the art?

- RQ2: What computer vision models are employed in research?

how is question related to SE?

- RQ3: What are the primary application areas of generative AI within computer vision?

integrations of computer vision models and techniques maybe?

- RQ4: What challenges are associated with computer vision in software engineering tasks

- RQ5: What are the primary performance metrics used to assess computer vision–based approaches across different software engineering tasks?

# 3 Datasets

This review considers studies sourced from established digital libraries and indexing platforms commonly used in Software Engineering research. The primary sources include IEEE Xplore, ACM Digital Library, SpringerLink, ScienceDirect, and Scopus. These repositories provide access to peer-reviewed journal articles and conference proceedings relevant to computer vision and software engineering.

The term *datasets* in this review refers to both:

You might want to consider google scholar as well.

- Empirical datasets used within the selected studies (e.g., screenshots, UI images, diagrams, logs, bug reports, or visual artifacts), and

- Scholarly datasets in the form of curated research publications retrieved from selected digital libraries.

Both publicly available and proprietary datasets reported in the studies are considered, provided they are used to evaluate computer vision–based approaches in software engineering tasks.

# 4 Keywords and Search Strings

## 4.1 Keywords

The following keywords are used to construct the search strategy:

- Computer Vision
- Generative AI
- Deep Learning not necessary CV related
- Software Engineering
- Software Development Life Cycle (SDLC)
- Software Development
- Software Testing
- Software Maintenance
- Program Analysis
- Performance Metrics
- User Interface Analysis UI is not part of SDLC?
- Challenges Too vague

## 4.2 Search Strings

Search strings are formed by combining computer vision–related terms with software engineering terms. Example search strings include: I beileve we have a single search string so you need to use OR

- (”computer vision” AND ”software engineering”)
- (”computer vision” AND ”software development”)
- (”computer vision” AND ”software development life cycle”)
- (”computer vision” AND ”software testing”)
- (”computer vision” AND ”software maintenance”)
- (”computer vision” AND ”program analysis”)
- (”computer vision” AND ”user interface analysis”)
- (”deep learning” AND ”computer vision” AND ”software engineering”)
- (”generative AI” AND ”computer vision” AND ”software engineering”)

- (”computer vision” AND ”software engineering” AND ”performance metrics”)
- (”computer vision” AND ”software engineering” AND challenges)
- (”computer vision” AND ”software engineering” AND limitations)
- (”generative AI” AND ”software development”)
- (”computer vision” AND ”software engineering”) AND (”performance metrics” OR ”accuracy” OR ”precision” OR ”recall”)

# 5 Inclusion Criteria

Studies are included in the review if they meet all of the following criteria:

- The study is published in a peer-reviewed journal, conference, or workshop. usually we dont consider workshops
- The study was published in the last ten years.
- The study focuses on the application of computer vision techniques within software engineering.

which specific keywords?

- The research addresses specific SDLC tasks.
- Provide sufficient methodological detail and empirical evaluation

# 6 Exclusion Criteria

short paper and other SLRs must be excluded, studies must be in top ranked venue and fully accessible online.

Studies are excluded from the review if they meet any of the following conditions:

- Not written in English
- Not aligned with the defined research questions (RQ1–RQ5) these are keyword related keep one as IC or EC
- Do not focus on the application of computer vision techniques within software engineering.
- Duplicate or redundant Redundant studies that present substantially similar results to the already included work
- Do not involve computer or machine vision techniques
- Grey literature such as blogs, tutorials, editorials, or non-peer-reviewed reports